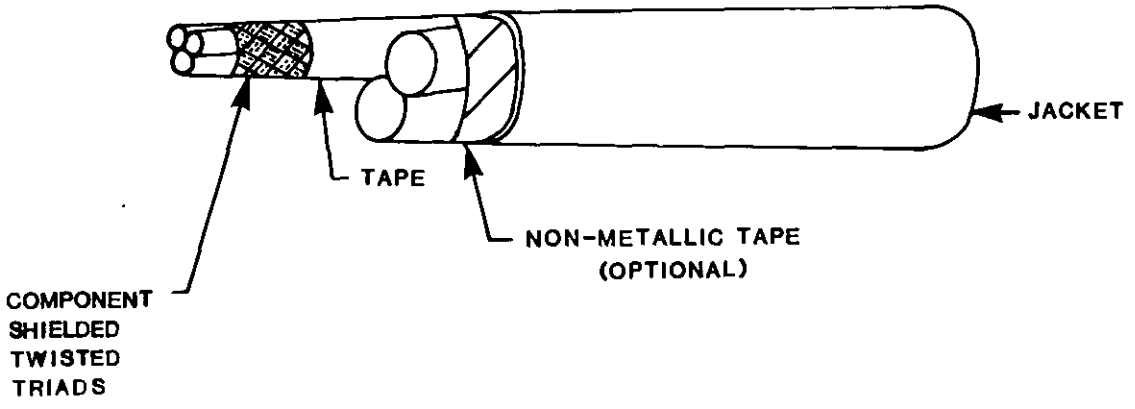
MILITARY SPECIFICATION SHEET

(A) CABLE, ELECTRICAL, 600 VOLTS, LIGHTWEIGHT,
MULTI-TRIAD, AWG 18, TYPE 3XSW
(INCLUDING VARIATION TYPES 3XSOW AND 3XSWA)

This military specification has been prepared by the Naval Sea Systems Command based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-C-24640. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24640.



SH 12765

COMPONENT SHIELDED TRIAD:

First - Copper conductor, AWG 18, 7 stranding. Nominal diameter of stranding conductor 0.049 inch.
Second - Material requirements in accordance with MIL-W-81044/12 except the conductor and identification codes and methods shall be in accordance with MIL-C-24640. Finished wire diameter 0.062 inch minimum and 0.068 inch maximum.
Third - Shield shall be tin-coated copper braid with a coverage not less than 85 percent determined in accordance with MIL-C-27500.

FIGURE 1. Construction (watertight).

(A) denotes changes.

FSC 6145

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(A) TABLE I. Construction details - unarmored.

Type number	Number of triads	Jacket thickness (inch) (minimum)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max)
			(min)	(max)	
3XSW-3	3	0.041	0.472	0.508	202.0
3XSW-7	7	.044	.620	.668	387.0
3XSW-10	10	.047	.803	.865	613.0
3XSW-14	14	.047	.873	.941	770.0

(A) TABLE II. Construction details - armored.

Type number	Number of triads	Jacket thickness (inch) (minimum)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max)
			(min)	(max)	
3XSWA-3	3	0.041	0.522	0.558	254.8
3XSWA-7	7	.044	.670	.718	458.1
3XSWA-10	10	.047	.853	.915	701.1
3XSWA-14	14	.047	.923	.991	872.7

(A) TABLE III. Construction details - overall shield.

Type number	Number of triads	Diameter over shield (inches) (nom)	Jacket thickness (inch) (minimum)	Cable diameter (inches)		Cable weight (pounds per 1000 feet) (max)
				(min)	(max)	
3XSOW-3	3	0.398	0.044	0.519	0.559	271.0
3XSOW-7	7	.547	.044	.659	.711	472.0
3XSOW-10	10	.721	.047	.835	.901	702.0
3XSOW-14	14	.800	.047	.898	.968	860.0
3XSOW-19	19	.909	.050	1.010	1.090	1127.0
3XSOW-24	24	1.080	.057	1.200	1.300	1489.0

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REQUIREMENTS:

Qualification required.
Construction details: See tables I, II and III.

When this cable is manufactured with armor, the following additional requirements shall apply:

(a) Braided metal armor and paint shall be added.
(b) Cable surface marking will not be required.

When this cable is manufactured with an overall shield, the following additional requirements shall apply:

(a) An overall tin-coated copper braid shall be added.
(b) Braid coverage shall be not less than 85 percent determined in accordance with MIL-C-27500.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum	7.29
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Conductor to shield	2000
Shield to shield (type 3XSOW only)	500
Insulation resistance - megohms/1000 feet, minimum	5000
Conductor and shield continuity (type 3XSOW only)	No failure
Jacket flaws (type 3XSOW only)	No failure
Group A:	
Visual and dimensional	No failure
Capacitance:	
Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum	55
Capacitance unbalance - percent, maximum	15
Characteristic impedance - at 1 MHz, ohms	45 ± 5
Watertightness (see table XI of MIL-C-24640 for limits of water leakage)	No failure
Weight	No failure
Group B:	
Cold bending	No failure
Drip	Zero
Tear - pounds per inch thickness, minimum	35
Physicals (unaged)	
Jacket	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Attenuation - at 3 MHz, dB/100 feet, maximum	4
Gas flame (1 hour)	Not applicable

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EXAMINATION AND TESTS: (Continued)

	Requirements
Group C:	
Physicals (aged)	
Jacket	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Permanence of printing (conductor) - cycles, minimum	50
Permanence of printing (jacket) - cycles, minimum	250
Armor (type 3XSWA only) - conformance to material, construction and coverage	No failure
Shield - conformance to material, construction and coverage	No failure
Heat distortion - percent unaged, maximum	30
Cable filler removability	No failure
Shrinkage	No failure
Mutual inductance - dB/1000 feet, maximum	Not applicable
Surface transfer impedance (type 3XSOW only)	
Milliohms per meter, maximum	700
EMP response, dB, minimum	60
Pulse response time - ms/500 feet	
Rise time, maximum	Not applicable
Decay time, maximum	Not applicable

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Aging and compatibility (cable)	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation	No failure
Immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	---
Insulation	10

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QUALIFICATION INSPECTION: (Continued)

	Requirements
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer - (type A) hardness, minimum	80
Weathering	No failure
Fungus resistance	No failure
Abrasion resistance	No failure
Gas flame (3 hour)	Not applicable
Finished wire	Not applicable

Preparing activity:
Navy - SH
(Project 6145-N322-18)

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